Status	Finished
Started	Friday, 30 October 2026, 3:28 PM
Completed	Friday, 30 October 2026, 3:31 PM
Duration	2 mins 57 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What will the below Python code do?

```
set1={2,3}

set2={3,2}

set3={2,1}

if(set1==set2):

    print("yes")

else:

    print("no")

if(set1==set3):

    print("yes")

else:

    print("no")
```

a. No, No

b. "==" is not supported for set in Python

c. Yes, No ✓

d. Yes, Yes

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following Python code will create a set?

(i) set1=set((0,9,0))
(ii) set1=set([0,2,9])
(iii) set1={}

a. All of the above
b. ii
c. i,ii ✓
d. iii

Question **3**

Correct

Mark 1.00 out of 1.00

What is printed when the following code is run?

```
tup = ('30', '3', '2', '8')
print(sorted(tup,reverse = True))
```

a. ['2', '3', '30', '8']
b. ['2', '3', '8', '30']
c. ['30', '8', '3', '2']
d. ['8', '30', '3', '2'] ✓

Question **4**

Correct

Mark 1.00 out of 1.00

What will set1|set2 do?

```
If set1={"a","b",3}
set2={3,7}
```

a. A new set will be created with the unique elements of set1 and set2.
b. Elements of set1 will get appended to set2
c. A new set will be created with the elements of both set1 and set2 ✓
d. Elements of set2 will get appended to set1

Question **5**

Correct

Mark 1.00 out of 1.00

What will be the output of following Python code?

```
set1={2,5,3}

set2={3,1}

set3={}

set3=set1&set2

print(set3)
```

a. {2,5,3,1}

b. {2,5,1}

c. {}

d. {3} ✓

Question **6**

Correct

Mark 1.00 out of 1.00

A python tuple can be created without using any parentheses. (True/False)

a. False

b. True ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Choose the correct option.

a. In Python, a tuple can contain only integers as its elements.

b. In Python, a tuple can contain only strings as its elements.

c. In Python, a tuple can contain either string or integer but not both at a time.

d. In Python, a tuple can contain both integers and strings as its elements. ✓

Question **8**

Correct

Mark 1.00 out of 1.00

What is the output of the following set operation

```
sampleSet = {"Yellow", "Orange", "Black"}
sampleSet.update(["Blue", "Green", "Red"])
print(sampleSet)
```

a. TypeError: update() doesn't allow list as a argument.

b. {'Yellow', 'Orange', 'Red', 'Black', 'Green', 'Blue'} ✓

c. Name Error

d. {'Yellow', 'Orange', 'Black', ["Blue", "Green", "Red"]}

Question **9**

Correct

Mark 1.00 out of 1.00

```
What is the output of the following code?
```

```
aTuple = (10, 20, 30, 40, 50, 60, 70, 80)
print(aTuple[2:5], aTuple[:4], aTuple[3:])
```

a. (10, 20, 30, 40) (40, 50, 60, 70, 80)

b. (30, 40, 50) (10, 20, 30, 40)

c. (30, 40, 50) (10, 20, 30, 40) (40, 50, 60, 70, 80) ✓

d. (30, 40, 50)(40, 50, 60, 70, 80)

Question **10**

Correct

Mark 1.00 out of 1.00

What is the output of the given below program?

```
t1 = (1,2,3)
t2 = (4,5,6)
x = t1+t2
print(x)
```

- a. (1,2,3,3,2,1)
- b. (1,2,3)(4,5,6)
- c. Error
- d. (1,2,3,4,5,6) ✓

Question **11**

Correct

Mark 1.00 out of 1.00

What will be the output of below Python code?

```
tupl=("python","programming","Computer")
print(tupl[-3:0])
```

- a. () ✓
- b. Computer
- c. (Computer)
- d. Error

Question **12**

Correct

Mark 1.00 out of 1.00

What is the output of the given below program?

```
t = (58, 47, 36, 25, 14, 3)
x = t[2:-1]
print(x)
```

a. Error

b. (58,47,36,25)

c. (36, 25, 14) ✓

d. (3,14,25)

Question **13**

Correct

Mark 1.00 out of 1.00

What will be the output of the below Python code?

```
t1=(55,12,78,64,25)
t1.pop(12)
print(tuple1)
```

a. Error ✓

b. (55,78,64,25)

c. (12)

d. 12

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following options will not result in an error when performed on tuples in Python where tupl=(5,2,7,0,3)?

a. tupl.append(2)

b. tupl.sort()

c. tupl[1]=2

d. tupl1=tupl+tupl ✓

Question **15**

Correct

Mark 1.00 out of 1.00

Find the output of the given Python program?

```
t1 = (55, 44, 33, 22, 11)
x = [t1[i] for i in range(0, len(t1), 2)]
print(x)
```

a. ([55,33,11])

b. (55,33,11)

c. [(55,33,11)]

d. [55, 33, 11] ✓

Status	Finished
Started	Friday, 30 October 2026, 2:52 PM
Completed	Friday, 30 October 2026, 3:00 PM
Duration	7 mins 57 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

The **DNA sequence** is composed of a series of nucleotides abbreviated as `'A'`, `'C'`, `'G'`, and `'T'`.

- For example, `"ACGAATTCCG"` is a **DNA sequence**.

When studying **DNA**, it is useful to identify repeated sequences within the DNA.

Given a string `s` that represents a **DNA sequence**, return all the `10`**-letter-long** sequences (substrings) that occur more than once in a DNA molecule. You may return the answer in **any order**.

Example 1:

```
Input: s = "AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT"
Output: ["AAAAACCCCC","CCCCCAAAAA"]
```

Example 2:

```
Input: s = "AAAAAAAAAAAAA"
Output: ["AAAAAAAAAA"]
```

For example:

Input	Result
AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT	AAAAACCCCC CCCCCAAAAA

Answer: (penalty regime: 0 %)

```
s = input().strip()

seen = set()
repeated = []

for i in range(len(s) - 9):
    sequence = s[i:i+10]
    if sequence in seen and sequence not in repeated:
        repeated.append(sequence)
    else:
        seen.add(sequence)

if repeated:
    print(*repeated, sep="\n")
else:
    print("No repeated sequences")
```

	Input	Expected	Got	
✓	AAAAACCCCCAAAAACCCCCCAAAAAGGGTTT	AAAAACCCCC CCCCCAAAAA	AAAAACCCCC CCCCCAAAAA	✓

	Input	Expected	Got	
✓	AAAAAAAAAAAA	AAAAAAAAAA	AAAAAAAAAA	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **2**

Correct

Mark 1.00 out of 1.00

Check if a set is a subset of another set.

Example:

Sample Input1:

mango apple

mango orange

mango

output1:

yes

set3 is subset of set1 and set2

```
input2:
```

```
mango orange
```

```
banana orange
```

```
grapes
```

```
output2:
```

no

For example:

Test	Input	Result
1	mango apple mango orange mango	yes set3 is subset of set1 and set2
2	mango orange banana orange grapes	No

Answer: (penalty regime: 0 %)

```
set1 = set(input().split())
set2 = set(input().split())
set3 = set(input().split())

if set3.issubset(set1) and set3.issubset(set2):
    print("yes")
    print("set3 is subset of set1 and set2")
else:
    print("No")
```

	Test	Input	Expected	Got	
✓	1	mango apple mango orange mango	yes set3 is subset of set1 and set2	yes set3 is subset of set1 and set2	✓
✓	2	mango orange banana orange grapes	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **3**

Correct

Mark 1.00 out of 1.00

Program to print all the distinct elements in an array. Distinct elements are nothing but the unique (non-duplicate) elements present in the given array.

Input Format:

First line take an Integer input from stdin which is array length n.

Second line take n Integers which is inputs of array.

Output Format:

Print the Distinct Elements in Array in single line which is space Separated

Example Input:

5

1 2 2 3 4

Output:

1 2 3 4

Example Input:

6

1 1 2 2 3 3

Output:

1 2 3

For example:

Input	Result
5 1 2 2 3 4	1 2 3 4

Answer: (penalty regime: 0 %)

```
n = int(input())

arr = []
for i in range(n):
    arr.append(int(input()))

result = []
for x in arr:
    if x not in result:
        result.append(x)

print(*result)
```

	Input	Expected	Got	
✓	5 1 2 2 3 4	1 2 3 4	1 2 3 4	✓
✓	6 1 1 2 2 3 3	1 2 3	1 2 3	✓
✓	5 11 22 11 22 11	11 22	11 22	✓
✓	10 1 2 3 4 5 1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **4**

Correct

Mark 1.00 out of 1.00

You are given an integer tuple `nums` containing distinct numbers. Your task is to perform a sequence of operations on this tuple until it becomes empty. The operations are defined as follows:

1. If the first element of the tuple has the smallest value in the entire tuple, remove it.
2. Otherwise, move the first element to the end of the tuple.

You need to return an integer denoting the number of operations required to make the tuple empty.

Constraints

- The input tuple `nums` contains distinct integers.
- The operations must be performed using tuples and sets to maintain immutability and efficiency.
- Your function should accept the tuple `nums` as input and return the total number of operations as an integer.

Example:

Input: nums = (3, 4, -1)
Output: 5

Explanation:
Operation 1: [3, 4, -1] -> First element is not the smallest, move to the end -> [4, -1, 3]
Operation 2: [4, -1, 3] -> First element is not the smallest, move to the end -> [-1, 3, 4]
Operation 3: [-1, 3, 4] -> First element is the smallest, remove it -> [3, 4]
Operation 4: [3, 4] -> First element is the smallest, remove it -> [4]
Operation 5: [4] -> First element is the smallest, remove it -> []
Total operations: 5

For example:

Test	Result
`print(count_operations((3, 4, -1)))`	5

Answer: (penalty regime: 0 %)

Reset answer

```
def count_operations(nums: tuple) -> int:
    operations = 0
    remaining = set(nums)
    nums = list(nums)

    while nums:
        first = nums[0]
        if first == min(remaining):
            remaining.remove(first)
            nums = nums[1:]
        else:
            nums = nums[1:] + [first]
        operations += 1

    return operations
```

	Test	Expected	Got	
✓	`print(count_operations((3, 4, -1)))`	5	5	✓
✓	`print(count_operations((1, 2, 3, 4, 5)))`	5	5	✓
✓	`print(count_operations((5, 4, 3, 2, 1)))`	15	15	✓
✓	`print(count_operations((42, )))`	1	1	✓
✓	`print(count_operations((-2, 3, -5, 4, 1)))`	11	11	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **5**

Correct

Mark 1.00 out of 1.00

Coders here is a simple task for you, Given string str. Your task is to check whether it is a binary string or not by using python set.

Examples:

Input: str = "01010101010"

Output: Yes

Input: str = "REC101"

Output: No

For example:

Input	Result
01010101010	Yes
010101 10101	No

Answer: (penalty regime: 0 %)

```
s = input().strip()

if set(s).issubset({'0', '1'}):
    print("Yes")
else:
    print("No")
```

	Input	Expected	Got	
✓	01010101010	Yes	Yes	✓
✓	REC123	No	No	✓
✓	010101 10101	No	No	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **6**

Correct

Mark 1.00 out of 1.00

Given a tuple and a positive integer k, the task is to find the count of distinct pairs in the tuple whose sum is equal to **K**.

Examples:

Input: t = (5, 6, 5, 7, 7, 8), K = 13
Output: 2
Explanation:
Pairs with sum K(= 13) are {(5, 8), (6, 7), (6, 7)}.
Therefore, distinct pairs with sum K(= 13) are { (5, 8), (6, 7) }.
Therefore, the required output is 2.

For example:

Input	Result
1,2,1,2,5 3	1
1,2 0	0

Answer: (penalty regime: 0 %)

```
import sys

# Read all input from stdin
input_data = sys.stdin.read().splitlines()

if input_data:
    # First line contains comma-separated numbers
    raw_nums = input_data[0].replace(',', ' ').split()
    t = tuple(map(int, raw_nums))

    # Second line contains K
    K = int(input_data[1].strip()) if len(input_data) > 1 else 0

    seen = set()
    pairs = set()

    for num in t:
        complement = K - num
        if complement in seen:
            pairs.add(tuple(sorted((num, complement))))
        seen.add(num)

    print(len(pairs))
```

	Input	Expected	Got	
✓	5,6,5,7,7,8 13	2	2	✓
✓	1,2,1,2,5 3	1	1	✓

	Input	Expected	Got	
✓	1,2 0	0	0	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **7**

Correct

Mark 1.00 out of 1.00

Given an array of integers `nums` containing `n + 1` integers where each integer is in the range `[1, n]` inclusive.There is only **one repeated number** in `nums`, return *this repeated number*. Solve the problem using set.

Example 1:

```
Input: nums = [1,3,4,2,2]
```

```
Output: 2
```

Example 2:

```
Input: nums = [3,1,3,4,2]
```

```
Output: 3
```

For example:

Input	Result
1 3 4 4 2	4

Answer: (penalty regime: 0 %)

```
nums = list(map(int, input().split()))

seen = set()

for num in nums:
    if num in seen:
        print(num)
        break
    seen.add(num)
```

	Input	Expected	Got	
✓	1 3 4 4 2	4	4	✓
✓	1 2 2 3 4 5 6 7	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **8**

Correct

Mark 1.00 out of 1.00

Write a program to eliminate the common elements in the given 2 arrays and print only the non-repeating elements and the total number of such non-repeating elements.

Input Format:

The first line contains space-separated values, denoting the size of the two arrays in integer format respectively.

The next two lines contain the space-separated integer arrays to be compared.

Sample Input:

5 4

1 2 8 6 5

2 6 8 10

Sample Output:

1 5 10

3

Sample Input:

5 5

1 2 3 4 5

1 2 3 4 5

Sample Output:

NO SUCH ELEMENTS

For example:

Input	Result
5 4 1 2 8 6 5 2 6 8 10	1 5 10 3
5 5 1 2 3 4 5 1 2 3 4 5	NO SUCH ELEMENTS

Answer: (penalty regime: 0 %)

```
N, M = map(int, input().split())

a = list(map(int, input().split()))
b = list(map(int, input().split()))

result = [x for x in a if x not in b] + [x for x in b if x not in a]

if result:
    print(*result)
    print(len(result))
else:
    print("NO SUCH ELEMENTS")
```

	Input	Expected	Got	
✓	5 4 1 2 8 6 5 2 6 8 10	1 5 10 3	1 5 10 3	✓
✓	3 3 10 10 10 10 11 12	11 12 2	11 12 2	✓
✓	5 5 1 2 3 4 5 1 2 3 4 5	NO SUCH ELEMENTS	NO SUCH ELEMENTS	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **9**

Correct

Mark 1.00 out of 1.00

There is a malfunctioning keyboard where some letter keys do not work. All other keys on the keyboard work properly.

Given a string text of words separated by a single space (no leading or trailing spaces) and a string brokenLetters of all distinct letter keys that are broken, return the number of words in text you can fully type using this keyboard.

Example 1:

Input: text = "hello world", brokenLetters = "ad"

Output:

1

Explanation: We cannot type "world" because the 'd' key is broken.

For example:

Input	Result
hello world ad	1
Faculty Upskilling in Python Programming ak	2

Answer: (penalty regime: 0 %)

```
import sys

lines = sys.stdin.read().splitlines()

text = lines[0].strip()
broken = lines[-1].strip().lower()

count = 0

for word in text.split():
    if not any(ch.lower() in broken for ch in word):
        count += 1

print(count)
```

	Input	Expected	Got	
✓	hello world ad	1	1	✓
✓	Welcome to REC e	1	1	✓
✓	Faculty Upskilling in Python Programming ak	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

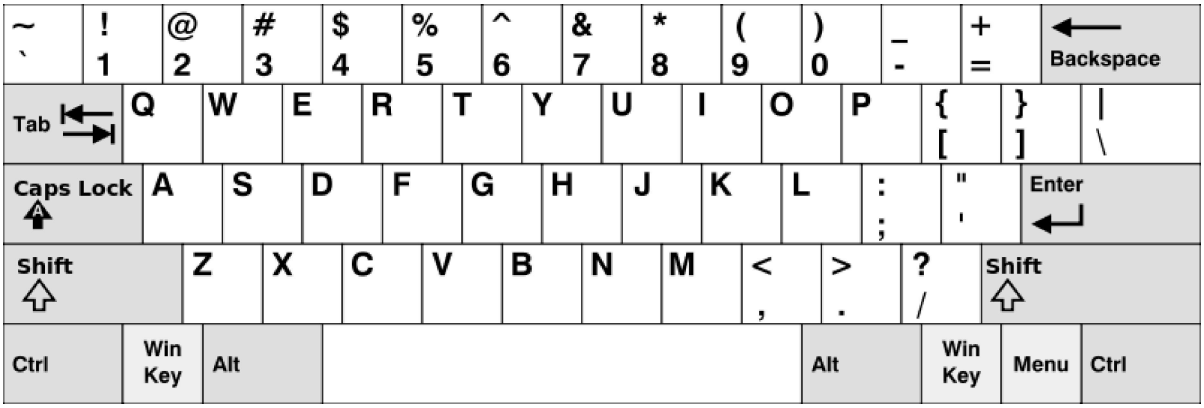
Correct

Mark 1.00 out of 1.00

Given an array of strings `words`, return *the words that can be typed using letters of the alphabet on only one row of American keyboard like the image below.*

In the **American keyboard**:

- the first row consists of the characters `"qwertyuiop"`,
- the second row consists of the characters `"asdfghjkl"`, and
- the third row consists of the characters `"zxcvbnm"`.



Example 1:

```
Input: words = ["Hello","Alaska","Dad","Peace"]
Output: ["Alaska","Dad"]
```

Example 2:

```
Input: words = ["omk"]
Output: []
```

Example 3:

```
Input: words = ["adsdf","sfd"]
Output: ["adsdf","sfd"]
```

For example:

Input	Result
4 Hello Alaska Dad Peace	Alaska Dad
2 adsfd afd	adsfd afd

Answer: (penalty regime: 0 %)

```
n = int(input())
words = [input().strip() for _ in range(n)]

row1 = set("qwertyuiop")
row2 = set("asdfghjkl")
```

```
row3 = set("zxcvbnm")

result = []

for word in words:
    letters = set(word.lower())
    if (letters.issubset(row1) or
        letters.issubset(row2) or
        letters.issubset(row3)):
        result.append(word)

if result:
    print(*result, sep="\n")
else:
    print("No words")
```

	Input	Expected	Got	
✓	4 Hello Alaska Dad Peace	Alaska Dad	Alaska Dad	✓
✓	1 omk	No words	No words	✓
✓	2 adsfd afd	adsfd afd	adsfd afd	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.